Macroeconomics III - Weekly notes

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2026

1 Week 1

1.1 Assumptions

- Time is discrete (i.e., not continuous) - only two moments matter, today and tomorrow. no time in between. only these two snapshots
- There is a representative agent. - One person represents entire economy. everyone is identical, thus studying one person means studying the entire economy.
- Two period model $t \in \{0, 1\}$ Captures the key tradeoff (consume now vs. later) without adding complexity. there is only today and tomorrow
- Agents discount the future at factor β - How much you value future consumption relative to today. example: $\beta = 0.96$ means 1 unit of consumption tomorrow = 0.96 units today
- The agent can save in bonds (a_1) which pay interest rate $R(= 1 + r)$
- Each period, agents receive an endowment y_t, which is known. Key: You can't change your income, only decide how to spend/save it
- The utility function is given by $u(c_t)$ - measures happines :-) consuming c_1.

  No general equilibrium yet
- Assume R is given exogenously ("Partial equilibrium"). You can't affect it—you're a price-taker.

1.2 Solving the Two-Period Problem

idea:

- Consume more today → less savings → less to consume tomorrow
- Consume less today → more savings → more to consume tomorrow

We want to maximize total utility across both periods:

$$\begin{aligned} \max_{c_0,c_1} \quad & U = u\left(c_0\right) + \beta u\left(c_1\right) \\ \text{subject to} \quad & c_0 + a_1 = y_0 \\ & c_1 = y_1 + (1+r)a_1 \end{aligned}$$

The two constraints are period 0 and 1 budget constraints. **problem**: We have 3 choice variables ($\mathrm{c}_0, \mathrm{c}_1, \mathrm{a}_1$) and 2 constraints. **Solution:** combine the two constraints (see written notes for calculations):

$$c_0 = y_0 - \frac{c_1 - y_1}{1+r}$$

This equaiton now basically says: "how much can i consume today if i want to consume c_1 tomorrow." the fraction on right side tells how much savings need to carry into period 1 to meet period 1 consumption demands. we can substitute the combined constraint into the utility function to obtain a simpler max problem:

$$\max_{c_1} U = u\left(y_0 - \frac{c_1 - y_1}{1+r}\right) + \beta u\left(c_1\right)$$

To find the maximum we take the derivative of U with respect to c_1 and set it equal to zero. after rearranging this gives us (see written notes for calcs):

$$\beta u'\left(c_1\right) = \frac{1}{1+r} \cdot u'\left(c_0\right)$$

The first order conditionis is the equation you get when you take the derivative of your objective function and set it equal to zero to find the maximum. When you're maximizing something (like utility), the FOC tells you where the peak is ie. where you can't improve by moving in either direction. The above equation can be re-arranged into the euler equation:

$$u'\left(c_0\right) = \beta(1+r)u'\left(c_1\right)$$

- **left side:** marginal utility of consuming one more unit today
- **right side:** marginal utility of consuming one more unit tomorrow adjusted for impatience β and interest $1+r$

The euler equation basically say: "at the optimum, the marginal benefit of consuming today equals the marginal benefit of saving and consuming tomorrow" If this weren't true,

you could make yourself better off by shifting consumption between periods → thus you have not found the optimal bundle. additionally we can rearrange like this:

$$\frac{u'(c_0)}{\beta u'(c_1)} = 1 + r$$

The left side is the marginal rate of substitution between consumption today and tomorrow. **The right side** is the market exchange rate between today and tomorrow (the interest rate). At the optimum, these must be equal.

To fully solve the problem we need:

- The Euler equation: $u'(c_0) = \beta(1 + r)u'(c_1)$ (This tells you the optimal ratio between c_0 and c_1)
- The budget constraint: $c_1 = y_1 + (1 + r)(y_0 - c_0)$ (tells you feasible combinations)

1.3 Utility function

crucial assumptions on what makes a good utility function. we need properties that make economic sense:

- Positive but Decreasing Marginal Utility. More consumption is always better. $u'(c) > 0$
- Decreasing marginal utility: $u''(c) < 0$. Each additional unit brings smaller increases in happiness. This is why indifference curves are convex (curved inwards).
- The limit condition: $\lim_{c\to\infty} u'(c) = 0$ As you consume more and more, the marginal benefit approaches zero but will never be negative
- Consumption is Essential $\lim_{c\to 0} u'(c) = \infty$. As consumption approaches zero (starvation), marginal utility explodes to infinity

1.3.1 Constant Relative Risk Aversion (CRRA)

The formula:

$$u(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma}; \quad \sigma > 0; \sigma \neq 1$$

The workhorse utility function in economics adhering to all of the above properties. Taking the derivative:

$$u'(c) = \frac{d}{dc}\left[\frac{c^{1-\sigma} - 1}{1 - \sigma}\right] = \frac{(1 - \sigma)c^{-\sigma}}{1 - \sigma} = c^{-\sigma}$$

we see the marginal utility is just:

$$u'(c) = c^{-\sigma}$$

The Parameter σ (Sigma)

- Coefficient of Relative Risk Aversion: How much you dislike uncertainty/variability in consumption. High risk aversion → hate fluctuations, prefer smooth consumption.
- Intertemporal Elasticity of Substitution = $1/\sigma$ This measures how willing you are to shift consumption between periods when the interest rate changes.
- As $\sigma \to \infty$, Leontief Preferences

2 Week 2

Two periods are not enough. We introduce an arbitrary, but finite horizon:

Preferences

$$U = u(c_0) + \beta u(c_1) + \beta^2 u(c_2) + \cdots = \sum_{t=0}^{T} \beta^t u(c_t)$$

Dynamic budget constraints:

$$\begin{aligned} c_0 + a_1 &= y_0 + (1 + r_0) a_0 \\ c_1 + a_2 &= y_1 + (1 + r_1) a_1 \\ c_t + a_{t+1} &= y_t + (1 + r_t) a_t \\ a_{T+1} &\geq 0 \end{aligned}$$

problem components the problem then becomes:

$$\max_{c_t, a_{t+1} \forall t} U = \sum_{t=0}^{T} \beta^t u(c_t)$$

subject to:

$$c_t + a_{t+1} = y_t + (1 + r_t) a_t; a_{T+1} \geq 0; a_0 = 0$$

bottom line: we're solving a household's consumption problem over multiple periods (from time 0 to time T). The household wants to maximize lifetime happiness by choosing how much to consume (c_t) and save (a_{t+1}) each period. What limits choices are the budget constraint:

$$c_t + a_{t+1} = y_t + (1 + r_t) a_t$$

this says: What you spend + what you save = what you earn + what you saved before (with interest). this problem can be setup and solved through **Lagrangian:**

$$\mathcal{L} = \sum_{t=0}^{T} \beta^t u(c_t) + \sum_{t=0}^{T} \lambda_t \left[y_t + (1 + r_t) a_t - c_t - a_{t+1} \right] + \mu a_{T+1}$$

Instead of solving T separate optimization problems (one for each period), the Lagrangian lets us solve everything simultaneously by:

- Taking derivatives with respect to all choice variables (c_t and a_{t+1} for each t)
- Setting these derivatives equal to zero
- This gives us the Euler equation and other optimality conditions

Euler equation is identical to 2-period model:

$$u'(c_t) = \beta(1 + r_{t+1})\, u'(c_{t+1})$$

Infinite horizon The household now lives forever (or thinks infinitely far ahead):

$$\sum_{t=0}^{T} \beta^t u(c_t) \to \sum_{t=0}^{\infty} \beta^t u(c_t)$$

This represents intergenerational altruism (parents care about children who care about their children, etc.)

no ponzi condition: Finite horizon had: $a_{T+1} \geq 0$ "cannot die with debt". Infinite horizon has:

$$\lim_{T\to\infty} \left(\prod_{t=0}^{T} \frac{1}{1 + r_t} \right) a_{T+1} \geq 0$$

This is the discount factor that converts future assets to present value. The present value of your assets far in the future cannot be negative

You can borrow, but you must eventually pay it back (at least in present value terms). You can't roll over debt forever without eventually paying it back. The No-Ponzi condition is the infinite-horizon equivalent of "you can't die with debt." It prevents unrealistic Ponzi schemes where you borrow forever without repaying, ensuring the optimization problem has a sensible solution.

2.1 Ramsey model

The Ramsey model is essentially the Solow growth model with smart consumers. Instead of assuming people save a fixed fraction of their income, the Ramsey model lets people choose how much to save based on what's best for them.

key players

- Households: Choose consumption (c_t) and saving (a_t)

- Firms: Use capital (K_t) and labor (L_t) to produce output using $Y_t = AK_t^{\alpha}L_t^{1-\alpha}$ Firms borrow capital from households and pays r_t per unit of capital.

key equations Model boils down to these two equations:

- Resource constraint (law of motion for capital): $k_{t+1} = (1-\delta)k_t + f(k_t) - c_t$ this is how capital evolves: Capital tomorrow = leftover capital today (after depreciation) + new production - what we consumed. This describes total capital in the economy.
- euler equation: $u'(c_t) = \beta(1 + r_{t+1} - \delta)u'(c_{t+1})$ This balances "Should I consume today or tomorrow?". $r_{t+1} = f'(k_{t+1})$ = the return on capital (what you earn by saving)

2.2 steady state

The steady state is simply the long-run resting point of the economy - where nothing is changing anymore. Both consumption c and capital k have stopped moving.

ss for capital accum. In steady state, $k_{t+1} = k_t = k$ (capital stops changing). So simply substitute that into resource constraint:

$$k = (1-\delta)k + f(k) - c$$

solve for c:

$$c = f(k) - \delta k$$

This says: steady-state consumption = total output minus what you need to replace worn-out capital.

ss for euler equation In steady state, $c_{t+1} = c_t = c$ (consumption stops changing too). Substitute:

$$u'(c) = \beta(1 + r - \delta)u'(c)$$

rearrange:

$$r = \frac{1}{\beta} + \delta - 1$$

From the firm's optimization (slide 18), the firm hires capital until its marginal product equals the rental rate: $r = f'(k)$ Substituting this into for r:

$$f'(k) = \frac{1}{\beta} + \delta - 1$$

this says: The marginal product of capital must exactly compensate households for waiting (impatience, captured by $1/\beta$) and for depreciation (δ).

3 Week 4 - OLG Model

The Ramsey model assumed people live forever. That's obviously unrealistic. The OLG model fixes this: people live for only **2** periods - young and old - and new generations are constantly being born. This creates a richer and more realistic economy, and raises the question: can the government actually improve things?

- Time is discrete (period 1, 2, 3...)
- Each person lives 2 periods: young (works, earns wage, saves) and old (retires, lives off savings)
- Population grows at rate **n**, so each generation is larger than the last: $L_{t+1} = (1+n)L_t$
- Key difference from Ramsey: no dynasties — you don't care about your kids' utility

3.1 Household problem

Similar to previous setups. Young people maximise lifetime utility:

$$\max U = u\left(c_{1,t}\right) + \beta \cdot u\left(c_{2,t+1}\right)$$

budget constraints:

- When young: $c_{1,t} + a_{t+1} = w_t \rightarrow$ you earn a wage, consume some, save the rest (a)
- When old: $c_{2,t+1} = (1 + r_{t+1}) \cdot a_{t+1} \rightarrow$ you live off your savings + interest

3.1.1 Euler equation

also very similar logic:

$$u'\left(c_1\right) = \beta(1+r) \cdot u'\left(c_2\right)$$

At the optimum, giving up a unit of consumption today (losing $u'\left(c_1\right)$) should equal the discounted gain from consuming more tomorrow.

3.2 Firms

Cobb-Douglas production: $Y = K^{\alpha}L^{1-\alpha}$ as always:

- $r = \alpha k^{\alpha-1}$ (return to capital)
- $w = (1-\alpha)k^{\alpha}$ (wage per worker)

3.3 Capital accum.

Young people's savings become next period's capital stock:

$$K_{t+1} = S_t \quad \Rightarrow \quad (1+n)k_{t+1} = s_t$$

With log utility + Cobb-Douglas, this gives the law of motion for capital:

$$(1+n)k_{t+1} = (1-\alpha)\frac{\beta}{1+\beta}k_t^{\alpha}$$

The Golden Rule capital level k_{gr} is the capital stock that maximises steady-state consumption for everyone. It's defined by:

$$f'(k_{gr}) = n$$

the return on one extra unit of capital (the marginal product) equals the population growth rate. This is the "sweet spot" — just the right amount of capital.

Core porblem: The economy can end up at $k > k_gr^*$ (dynamically inefficient - too much capital, too much saving). If the capital stock is so large that the cost of maintaining it exceeds what it produces, you're wasting resources. Everyone would be better off if the economy had less capital and people just consumed more instead. But here is the problem: Nobody individually has an incentive to save less, because if you save less, you'll be poor when old.

3.4 Role of government

- In the Ramsey model, there's one representative household → easy to evaluate welfare.
- In OLG, infinitely many households (different generations) live at the same time. Whose utility counts more? There's no clear answer.
- Instead, we ask: can the government achieve a Pareto improvement (make someone better off without hurting anyone)?

The government steps in to coordinate a redistribution that no individual can pull off alone to address the core problem explained above. The government is transferring money between generations — specifically, taking from the young and giving to the old (or equivalently, issuing bonds that the young buy, then paying them back when they're old using taxes on the next young generation).

3.4.1 government debt:

Here $g = 0$ - the government doesn't spend anything. Instead it just moves money between generations.

Period t:

- Government issues bonds B_{t+1} (borrows money)
- The young buy these bonds - this is part of their saving
- Government uses the proceeds to pay off old-period bonds + gives transfers to the current old

Period t+1

- The same young are now old
- Government taxes the **new** young generation
- Uses that tax revenue to pay back the bonds (with interest) to the now-old

people can't make this scheme themselves because the next generation isn't born yet. You can't write a contract with someone who doesn't exist. The government, because it's infinitely lived, can make and enforce that promise across generations.

Importance of population growths: The government's "trick" only works because there are more young people each period (population grows at rate n).
Each old person is owed repayment, but there are $(1 + n)$ young people per old person to tax. So the effective return on this scheme is n - the population growth rate.
The scheme is worth doing when $n > r$ (population growth return beats the market return on capital) — which is exactly the condition $f'(k^*) < n$, i.e. dynamic inefficiency.

4 Week 5

In the OLG model, the market equilibrium can end up with too much capital — the economy over-saves. This happens when the steady-state interest rate ends up being lower than the population growth rate n. this would be a symptom that there is so much capital in the economy, the return on invested capital is now lower than the population growth. That's the condition for dynamic inefficiency.

the planners deal: Every young person gives a small amount x to the current old generation.

- when you are young in period t, you give up x meaning you loose x amount of consumption. not initially cool.
- but when you are old in period $t+1$ the new young generation all give you x each back.
- because there are now more young people: $1+n$ for every old person, you earn a return of $x(1+n)$ on the inital x you gave up in period t

This planner's scheme beats the market whenever:

$$n > r^*$$

ie. exactly the dynamic inefficiency condition. The planner is replacing a low-return investment (capital) with a higher-return deal (intergenerational transfers). Empirically, most economists think real economies have $r > n$ most of the time, so dynamic inefficiency is more of a theoretical possibility than a practical concern.

4.1 Governments debt

In the baseline OLG model without debt, young people save and all savings become capital:

$$(1+n)k_{t+1} = s_t$$

But now the young have two assets to save in: capital and government bonds. Both pay the same return (competitive markets, no arbitrage), so:

$$(1+n)\left(k_{t+1} + b_{t+1}\right) = s_t$$

Total savings hasn't changed much, but now part of it goes to bonds instead of capital. This is crowding out — government debt absorbs private savings that would otherwise fund productive capital investment.

debt - good or bad here? This is where the dynamic inefficiency story connects. If the economy is dynamically inefficient ($r^* < n$, too much capital), then crowding out is actually a good thing - the economy was over-saving, and debt soaks up some of those excess savings. The debt is essentially doing the same job as the planner's transfer scheme from the previous slide. The young buy bonds (giving resources to the government), the government uses the proceeds to pay off old bondholders (transferring resources to the

old). It's an intergenerational transfer implemented through debt rather than direct taxation.

TLDR: The mechanism is: government debt competes with capital for the young's savings, reducing the capital stock. Whether that's desirable depends on whether the economy was over-saving to begin with.

4.2 Social security I: Fully funded

A fully funded pension is conceptually simple: the government **forces** you to save a specific amount d_t out of your paycheck, invests it, and gives it back to you with interest when you retire. The key word is "fully funded" - your own contributions are set aside and invested for *your own* retirement. Nobody else's money is involved.

the math

- When young (period t): Your wage is w_t. You choose private savings s_t, and the government takes a mandatory contribution d_t :

$$c_{1,t} = w_t - s_t - d_t$$

- The government invests your d_t in the capital market, earning the same return r_{t+1} you'd earn yourself. So your pension benefit is:

$$b_{t+1} = (1 + r_{t+1})\, d_t$$

- When old (period $t + 1$): You get your private savings back plus your pension:

$$c_{2,t+1} = \underbrace{(1 + r_{t+1})\, s_t}_{\text{private savings}} + \underbrace{(1 + r_{t+1})\, d_t}_{\text{pension}} = (1 + r_{t+1})\,(s_t + d_t)$$

Look at old-age consumption. It depends on $s_t + d_t$ - your total savings, regardless of how it's split. If the government forces you to save $d_t = 5$, you just reduce your private savings by 5 (set s_t lower by 5), and everything is identical to before.

so why implement this at all? The slides hint at this: it only matters if people are **not** rational optimizers. If people have short planning horizons, procrastinate on saving, or have behavioral biases, then forced saving d_t helps because they *wouldn't* have saved that amount voluntarily. But within the standard rational model, it's completely neutral.

4.3 Social security II: Pay-as-you-go

In the fully funded system, your contributions were invested and returned to *you*. Here, your contributions d_t go **directly to today's old people** - they're spent immediately, not invested. When you're old, the *next* generation pays for you.

the math The benifit formula:

$$b_t = (1+n)d_t$$

Each old person receives b_t. There are ($1+n$) young people per old person, each paying d_t. So the old person gets $(1+n)d_t$. This is just accounting - no investment, no interest rate, just redistribution.

when young

$$c_{1,t} + s_t = w_t - d_t$$

Same as before - you earn w_t, the government takes d_t, you choose how much of the rest to save privately.

when old

$$c_{2,t+1} = \underbrace{(1+r_{t+1})\, s_t}_{\text{return on private savings}} + \underbrace{(1+n)d_{t+1}}_{\text{pension from next generation's young}}$$

Notice the critical difference from the fully funded case. There, both components earned return ($1+r$). Here, private savings earns ($1+r$), but the pension component earns an implicit return of $(1+n)$ - population growth, not the interest rate.

connection to the planner's solution Remember the planner's trick: set up intergenerational transfers where the young give x to the old and get $(1+n)x$ back from the next generation. Pay-as-you-go is exactly that scheme.

If the economy is dynamically inefficient ($r^* < n$), the pay-as-you-go return ($1+n$) exceeds the market return ($1+r$). By choosing the right d_t, the government can:

- 1. Reduce private savings (young people save less because they're paying d_t and expecting a pension)
- 2. Lower the capital stock toward the golden rule k_{gr}
- 3. Give everyone a better deal - the pension return $(1+n)$ beats what capital was offering $(1+r)$

4.3.1 comparisons table

	Fully funded	Pay-as-you-go
Your d_t goes to	Investment (for you)	Current old
Return on d_t	$(1+r)$	$(1+n)$
Affects Euler equation?	No	No (still lump sum)
Affects lifetime wealth?	No	Yes (if $r \neq n$)
Crowds out capital?	No	Yes
Can fix dynamic inefficiency?	No	Yes

5 Endogenous Labor Supply

Up to now in the course, the models have assumed people just work — labor supply is fixed, not a choice. But that's clearly unrealistic. Americans work way more hours than, say, the French. And we all work less than people did 100 years ago. To explain these patterns, we need to let households choose how much to work.

new utility function

$$U = u(c) - v(h) \quad \text{s.t.} \quad c = wh$$

This says: happiness comes from two things pulling in opposite directions:

- $u(c)$ - utility from consumption. More consumption = more happiness. This is the same as before in the course.
- $v(h)$ - dis-utility from working h hours. More work = more misery. The key detail is that this increases *exponentially* — the 8th hour of work feels way worse than the 2nd hour.

The **minus sign** is doing important work: you gain happiness from consuming but lose happiness from working. Total utility is the net of these two forces.

The **budget constraint** $c = wh$: Your consumption equals your wage rate w times hours worked h. No savings here - you eat what you earn. And the only way to consume more is to work more, which hurts.

Time budget $1 = h + l$: You have one unit of time (think of it as 24 hours, just normalized to 1). You split it between work h and leisure l. This is just bookkeeping - if you know h, you know $l = 1 - h$.

Optimality condition The household picks h to maximize $U = u(wh) - v(h)$. Taking the derivative with respect to h and setting it to zero:

$$\frac{d}{dh}[u(wh) - v(h)] = u'(c) \cdot w - v'(h) = 0$$

Rearranging:

$$\frac{v'(h)}{u'(c)} = w$$

What this says in plain English: the marginal rate of substitution between leisure and consumption equals the wage. In other words, the *extra misery* from one more hour of work (numerator), measured relative to how much you value the *extra consumption* that hour buys (denominator), must equal the wage w.

6 TODO

6.1 The many elasticities of labor supply

6.2 Labor demand

6.3 Risk